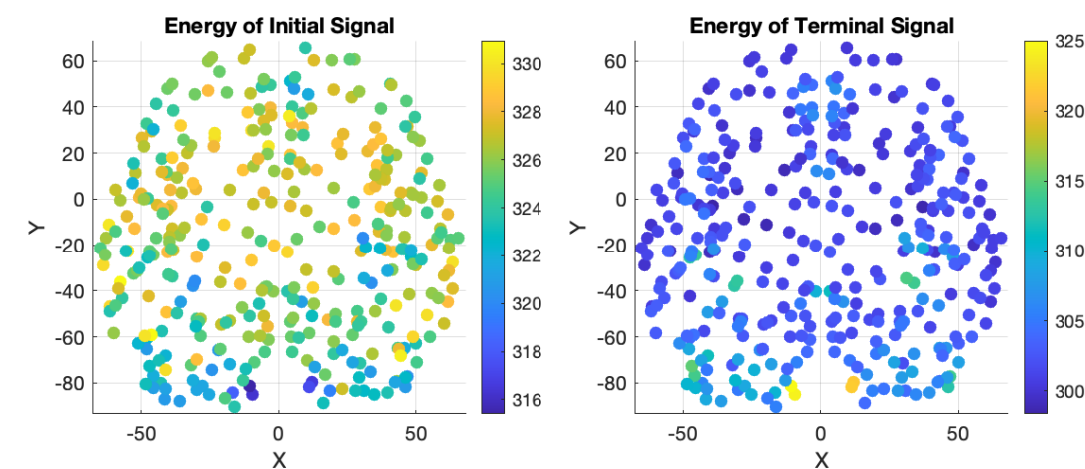


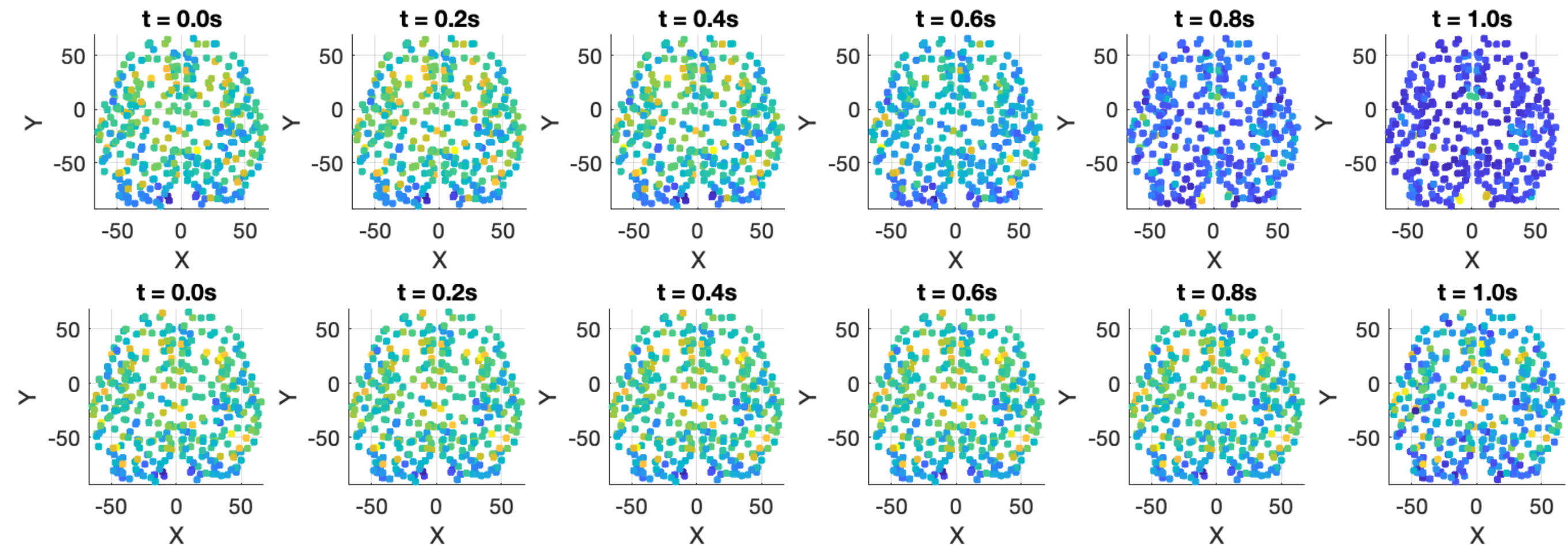
Topological signal matching

- Brain fMRI signals



liberal

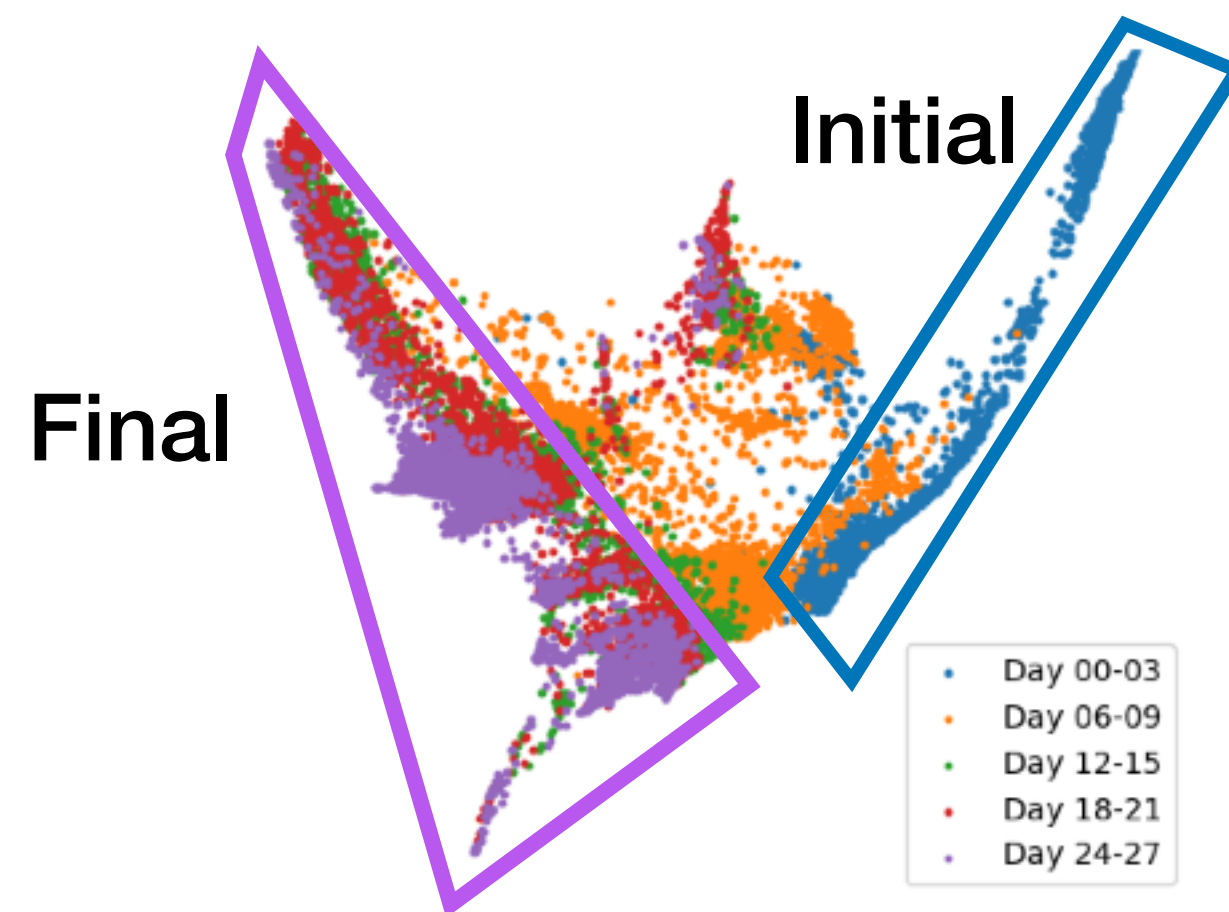
aligned



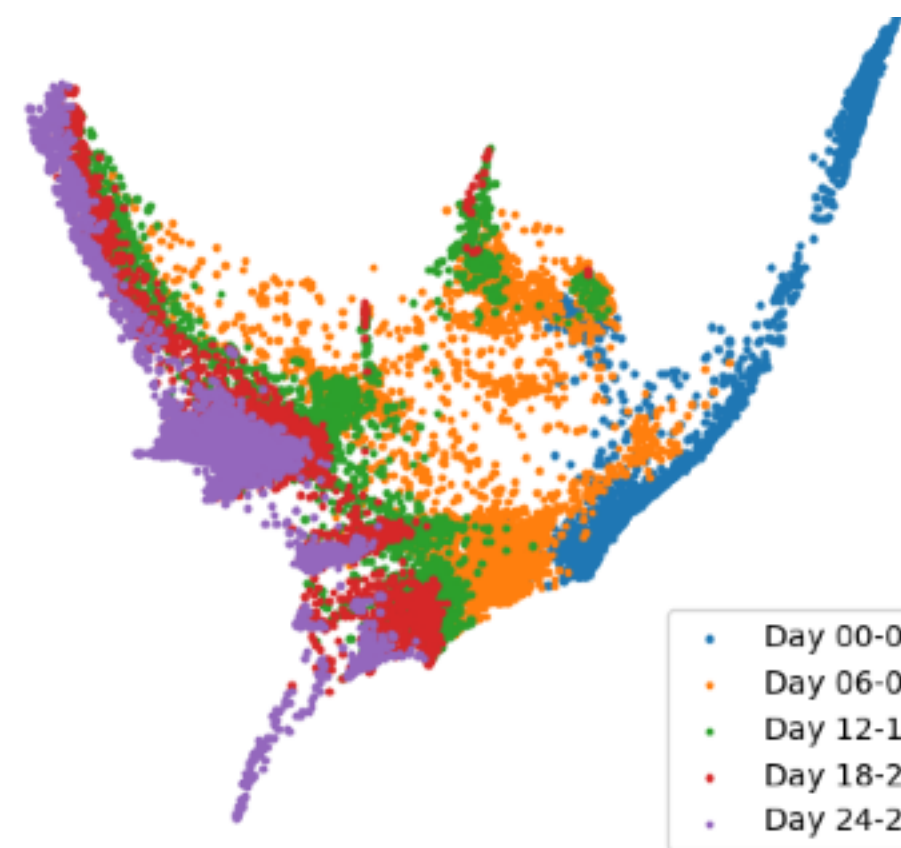
TSB-VE

SB-VE

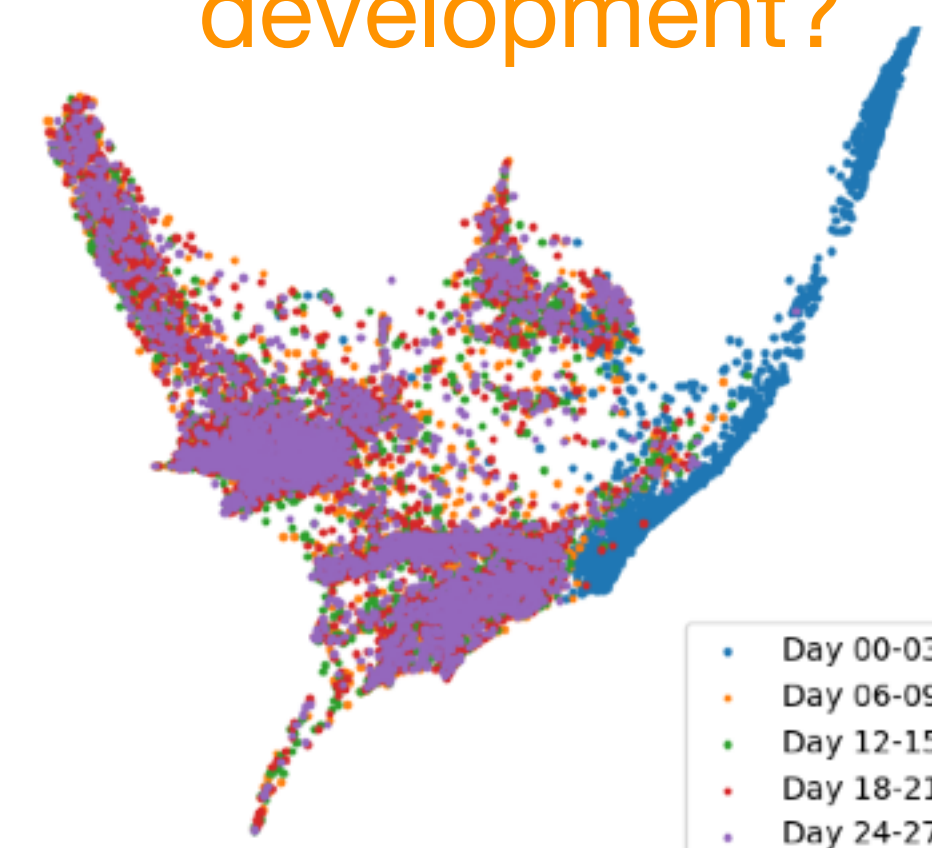
- Single-cell dynamics [Moon 2019, Tong 2023, 2024]



Groundtruth



TSB-BM



SB-BM

Potentials for chronic disease development?

Very similar work: geometric trajectory generating

- Modeling the geometric coordinates given node features and edge set

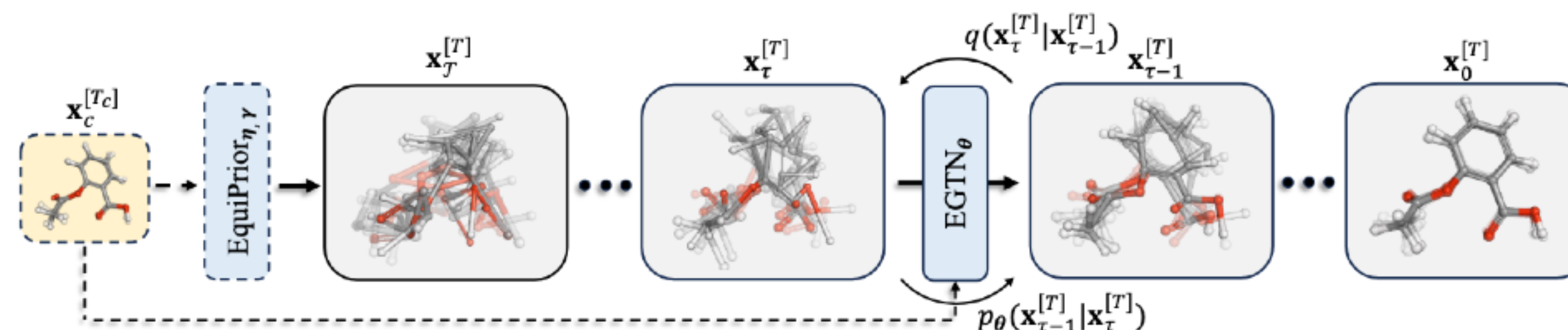
$p(\mathbf{x}^{[0,1,\dots]}, \mathbf{h}, \mathbb{E})$ (**But equivariance required**) (Han et al. 2024 neurips)

$\mathbf{x}$: node 3D coordinates

- Policy: Equivariant GNN + temporal-level feature attention

- If we view the coordinates as data on nodes, this is like our setting

$\mathbb{E}$: edge set, molecule topology



Diffusion model fashion

- If prior is SE3-inv, transition kernel is SE3-eqv., then marginals are SE3-inv
- Another Referred as Learning graph-dependent probability distribution $p_G(\mathbf{x})$

- Jing et al. 2023. Eigenfold: generative protein structure prediction with diffusion models.

- Reference process: $dx = -Hx dt + dw$, where $-Hx = \nabla E = \nabla(x^T Hx)$

- prior: $p(x_T) \propto \exp(-0.5x^T Hx) = \exp(-0.5 \sum_{i,j \in \mathbb{E}} |x_i - x_j|^2)$

Graph GP

